

RESEARCH ON FLIGHT PATH MEASUREMENT METHOD BASED ON MONOCULAR PHOTOGRAPHY AND INERTIAL DATA FUSION

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Abstract

In accordance with ICAO DOC 9501 Volume I, external noise certification flight tests typically employ a DGPS tracking system to measure aircraft trajectory data, with regulatory requirements for verifying the accuracy of the DGPS system. This paper proposes a method for aircraft trajectory measurement based on Monocular photography, investigates the Monocular photography measurement model and the unified coordinate system algorithm, and demonstrates the feasibility of using aircraft inertial navigation data to correct the three-dimensional position of the aircraft. The method was applied in actual flight tests to measure the trajectories of the aircraft at four different altitudes and compare the data. The results show that the deviation between Monocular photography-based aircraft trajectory measurement and DGPS system data is less than 1.5 meters. This verifies the accuracy of the DGPS tracking system as required by the regulations, providing significant evidence for the application of Monocular photography measurement technology in external noise testing.

1 Introduction

According to the requirements of ICAO DOC 9501 Volume I, in the subject of noise flight tests for civil aircraft, the basis for assessing aircraft noise is the Effective Perceived Noise Level (EPNL). In the calculation of EPNL parameters, it is necessary to measure the aircraft's trajectory and obtain the Pure-tone Corrected Perceived Noise Level (PNLT) by integrating the pure-tone corrected perceived noise level over time, based on the aircraft's altitude, flight trajectory, and airspeed. Generally, the trajectory data of the aircraft, such as longitude, latitude, and altitude, are included in the aircraft's onboard data bus. However, regulations explicitly require the use of devices external to the aircraft to obtain aircraft trajectory data. Therefore, precise and continuous measurement of aircraft trajectory parameters must be achieved through the installation of independent onboard test systems or external measurement systems. Currently, there are two typical methods for trajectory measurement: external optoelectronic measurement and installation of DGPS (Differential Global Positioning System) measurement. Among them, the external optoelectronic measurement methods mainly include radar tracking measurement, optoelectronic inertial measurement, and photogrammetry.

Radar tracking measurement utilizes radar-emitted electromagnetic waves to measure parameters such as distance, azimuth, and elevation angle of the target through receiving echo signals. The advantages of radar tracking measurement include high detection accuracy, wide detection range, good stability, and minimal influence from weather conditions, with high technological maturity. However, since

most large civil aircraft flight tests are conducted at civilian airports, the construction of radar measurement systems meeting measurement requirements entails high costs, long cycles, and significant expenses. Therefore, radar tracking is not suitable for trajectory measurement in large civil aircraft flight tests.

Optoelectronic inertial measurement usually requires the deployment of two optoelectronic inertial systems at both ends of the airport runway, utilizing triangulation to track and measure the aircraft's trajectory. The advantage of this method lies in the minimal impact on the aircraft's configuration, as it only requires the installation of targets or multi-face reflectors on the aircraft. However, the site selection, deployment, calibration, and operation of optoelectronic inertial systems are relatively complex and susceptible to factors such as lighting conditions and visibility. In terms of measurement range, the optimal area for optoelectronic inertial double-station triangulation measurement is between the two stations, and the measurement accuracy outside this optimal area will be significantly reduced, making it unsuitable for measuring the entire flight trajectory of the aircraft.

Photogrammetry typically employs high-speed cameras to capture images of the aircraft's flight process. By extracting key information from the images and calculating it in a pre-established coordinate system, the physical coordinate positions of the aircraft can be obtained, thus measuring the aircraft's trajectory. Compared to optoelectronic inertial measurement, photogrammetry has a simpler system architecture and lower deployment requirements. With

advancements in high-speed photography and image recognition technologies, photogrammetry has greatly improved in terms of sampling rate, image quality, and measurement accuracy. Currently, photogrammetry plays an important role in measuring aircraft position, take-off speed, glide angle, sink rate, and other flight parameters, serving as a significant complement to traditional optoelectronic inertial measurement and radar tracking methods. In the future, by increasing the number of cameras, optimizing coordinate system establishment and calibration methods, improving target recognition methods, and optimizing image processing algorithms, the measurement range and accuracy of photogrammetry can be further expanded, showing promising prospects.

DGPS, as a new generation of satellite navigation and positioning systems, has been widely used in the measurement field due to its global coverage, all-weather capability, high accuracy, and cost-effectiveness. In flight tests, DGPS technology can accurately measure aircraft trajectory, speed, and other flight state parameters. Its advantages lie in high measurement accuracy. Currently used DGPS measurement systems typically achieve dynamic positioning accuracy better than $2 \text{ cm} \pm 1 \text{ ppm}$ within a baseline length of 50-100 km. The disadvantage of DGPS measurement methods lies in the high cost of aircraft modification and the low refresh rate of GPS data, which cannot obtain high-frequency trajectory data.

According to the requirements of 3.2.1.3.4 in ICAO DOC 9501 Volume I, the accuracy of the DGPS trajectory measurement system used in flight tests needs to be verified, which should include comparing DGPS measurement data with aircraft trajectory data obtained by another method. This paper will focus on the photogrammetry method based on a monocular camera model, which innovatively uses inertial navigation data correction to improve trajectory measurement accuracy, and validates the reliability of DGPS-measured trajectories by comparing them with DGPS measurement results.

2 Measuring Principle

2.1 Camera Imaging Model

In computer vision, the projection of any object in three-dimensional space onto the corresponding image plane can be represented by a geometric model, and it is this geometric relationship that relates the two-dimensional coordinates of an image to the three-dimensional coordinates in space, which is commonly referred to as the camera imaging model.

The ideal camera imaging model is the linear camera model shown in Figure 1. Based on the imaging principle of the camera, the linear camera model uses the projection transformation to image a point P in space on the camera image plane, and the imaging point p is a linear relationship, that is, the space point P and the camera photocentre O, imaging point p three-point co-linear.

The following relationship can be obtained from Figure 1:

$$\begin{cases} x = f \frac{X}{Z} \\ y = f \frac{Y}{Z} \end{cases} \quad (1)$$

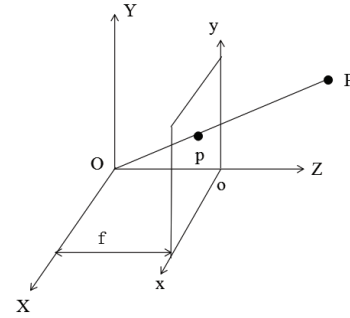


Fig. 1 Linear camera imaging model.

where (x, y) are the coordinates of point p under the image coordinate system, (X, Y, Z) are the coordinates of point P under the camera coordinate system, and f is the focal length.

Indeed, the standard perspective projection model deviates from linearity due to optical distortions inherent in lens systems. Real-world lenses are not ideal and exhibit various degrees of aberration, rendering the standard perspective projection model non-linear. However, in certain scenarios where the camera's field of view is narrow, the object distance remains relatively constant, or the object size is small relative to its distance from the camera, the model can be effectively simplified to a linear one. In light of the specific application context addressed in this paper, the subsequent theoretical discussion proceeds under the assumption of this simplified linear projection model.

2.2 Coordinate Measurement in the Camera Coordinate System

Under ideal conditions, assuming that the aircraft is flying level at a fixed altitude, according to the linear imaging model, the ratio of the aircraft altitude and the camera focal length is equal to the ratio of the actual length of the fixed scale on the aircraft and the length of its projection on the image. This is shown in Figure 2.

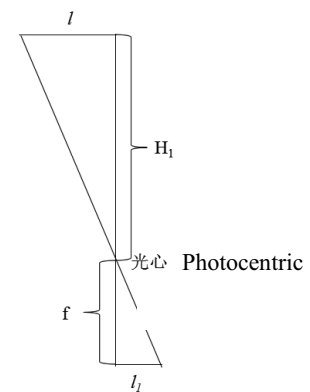


Fig. 2 Relationship between aircraft altitude, camera focal length, scale length, and scale projection.

where l is the actual length of the ruler on the aircraft, and the projection length of the ruler on the screen is l_1 when the aircraft is flying at an altitude of H_1 . Assuming a camera focal length of f , according to the linear imaging model, we can derive:

$$\frac{l}{l_1} = \frac{H_1}{f} \quad (2)$$

By utilizing the formula above, we can derive:

$$H_1 = \frac{fl}{l_1} \quad (3)$$

The size of the aircraft itself cannot be ignored, if we want to accurately solve the 3D spatial position of the aircraft, a feature point on the aircraft should be selected as a reference point, the coordinates of the reference point in the image coordinate system and the camera coordinate system under the coordinate relationship is shown in the following equation:

$$\frac{x}{x_c} = \frac{y}{y_c} = \frac{f}{z_c} = \frac{f}{H_1} = \frac{l}{l_1} \quad (4)$$

Where (x, y) represents the horizontal and vertical coordinates of point pp in the image under the image coordinate system, and (x_c, y_c, z_c) are the 3D coordinates of point PP under the camera coordinate system. Thus:

$$\begin{cases} x_c = \frac{x l_1}{l} \\ y_c = \frac{y l_1}{l} \\ z_c = H_1 \end{cases} \quad (5)$$

2.3 Harmonisation of Coordinate Systems

Since the camera coordinate system is a local reference coordinate system and the DGPS measurement result is the coordinate value with the WGS-84 geocentric coordinate system as the reference system, if the comparison of DGPS track data is carried out, it is firstly necessary to unify the two coordinate systems by converting the camera coordinate system to the WGS-84 geocentric coordinate system.

The conversion relationship between the two coordinate systems is as follows:

$$\mathbf{X}_G = \mathbf{A}_{\lambda_0} \mathbf{A}_{\varphi_0} \mathbf{X} + \mathbf{X}_{G_0} \quad (6)$$

Where: $\mathbf{X}_G = [X_G \ Y_G \ Z_G]^T$ is the coordinates of the measured target in the WGS-84 geocentric right-angle coordinate system, $\mathbf{A}_{\lambda_0}, \mathbf{A}_{\varphi_0}$ are the matrices corresponding to the astronomical latitude and astronomical longitude of the camera's optical centre point O, $\mathbf{X} = [x_c \ y_c \ z_c]^T$ is the

coordinate value of the target measured by the camera in the camera's coordinate system, and $\mathbf{X}_{G_0} = [X_{G_0} \ Y_{G_0} \ Z_{G_0}]^T$ is the coordinates of the camera's optical centre point O in the WGS-84 geocentric right-angle coordinate system.

To compare the DGPS measurements, it is also necessary to convert under the WGS-84 geocentric right-angle coordinate system to the WGS-84 geodetic coordinate system, and the conversion relationship is as follows:

$$X_0 = (N_0 + H_0) \cos B_0 \cos L_0 \quad (7)$$

$$Y_0 = (N_0 + H_0) \cos B_0 \sin L_0 \quad (8)$$

$$Z_0 = [N_0(1 - e^2) + H_0] \sin B_0 \quad (9)$$

$$N_0 = \frac{a}{(1 - e^2 \sin^2 B_0)^{1/2}} \quad (10)$$

$$e^2 = 1 - (b/a)^2 \quad (11)$$

Where X_0, Y_0, Z_0 are the coordinates of the measured target point under the WGS-84 Cartesian Coordinate System, L_0, B_0, H_0 are the geodetic longitude, latitude, and ellipsoidal height of the measured target point, N_0 is the radius of curvature of the dyadic circle passing over the measured target point, e is the first eccentricity of the WGS-84 ellipsoid, $a=6378137\text{m}$ is the long half-axis of the ellipsoid, $b=6356752.31414\text{m}$ is the short semi-axis of the ellipsoid.

Up to this point, the photogrammetric measurements and DGPS measurements have been converted to the same coordinate system, and the measurements of the two methods can be compared subsequently.

3 Image Matching Recognition Algorithm

After acquiring image data collected during the flight process, these images are utilized for target recognition and feature point extraction. In this study, the circular black and yellow cross-crossing target affixed to the aircraft is selected as the aircraft's feature region, as depicted in Fig. 3. Due to its high stability, adaptability to continuous changes in target traits and size, fast computation speed, and strong anti-interference performance, the MeanShift algorithm is employed for feature localization and tracking.

Following target identification and tracking in the image, the Harris corner detection method is applied to identify the corner point at the centre of the target as the feature point for aircraft positioning, providing the image coordinates of the corner point. The Harris corner detection method primarily utilizes the autocorrelation matrix of the point being measured to judge the corner point. Essentially, it employs a window of appropriate size on the image, moves it in any direction, and determines whether this point is a corner point based on the change in grey value of the pixel before and after movement. This method is utilized to ascertain whether the point qualifies as a corner point.

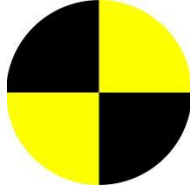


Fig. 3 Schematic of fuselage target.

4 Inertial Navigation Data Correction Methods

4.1 Altitude Correction Based on Inertial Navigation

During the actual flight process, as the aircraft traverses the camera's field of view, achieving ideal level flight isn't always feasible. Consequently, when designing and selecting the target scale, it's essential to consider the impact of the aircraft's attitude on its altitude determination. Moreover, corrections to the measurement of the aircraft's altitude should be made using inertial guidance data. The positioning of the designed target scale on the aircraft is illustrated in Figure 4.



Fig. 4 Fuselage target location.

The four targets situated on the belly of the aircraft can serve as two sets of reference points. The distance between the two targets positioned at the nose and tail of the aircraft constitutes one reference, while the separation between the two targets on the left and right wings forms another reference. It's evident that the pitch attitude of the aircraft doesn't impact the projection of the wing scale on the screen, and similarly, the roll attitude of the aircraft doesn't affect the projection of the belly scale on the screen. However, if the aircraft experiences both roll and pitch, one of the scales must be corrected to enhance the accuracy of the aircraft altitude measurement.

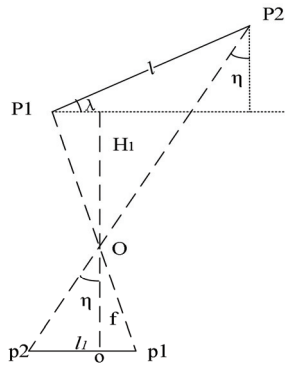


Fig. 5 Wing fiducials and their projected positions.

For instance, considering the correction of the wing scale, if the aircraft exhibits a roll angle denoted by λ , the correction algorithm proceeds as follows. In Fig. 5, the spatial positions of the two target scale centre points on the wing are denoted as P1 and P2, while their projections onto the image are represented by p1 and p2. Here, O signifies the optical centre point, and o denotes the image centre point.

If both wing targets are positioned to the right of the image centre or distributed on either side of the centre, the following relationship applies:

$$\begin{cases} \tan \eta = \frac{\sqrt{x_{p_2}^2 + y_{p_2}^2}}{f} \\ \frac{H_1}{f} = \frac{l \cos \lambda - l \sin \lambda \tan \eta}{l_1} \end{cases} \quad (12)$$

If both targets of the wing are to the left of the centre of the image, the following relation is given:

$$\begin{cases} \tan \eta = \frac{\sqrt{x_{p_2}^2 + y_{p_2}^2}}{f} \\ \frac{H_1}{f} = \frac{l \cos \lambda + l \sin \lambda \tan \eta}{l_1} \end{cases} \quad (13)$$

where λ is the roll angle of the aircraft, η is the auxiliary angle, and (x_{p_2}, y_{p_2}) are the image coordinates of point p_2 . H_1 is the height of point p_1 , and f is the lens focal length. Where l is the actual length of the ruler on the aircraft and l_1 is the projected length of the ruler on the screen.

4.2 Position Correction Based on Inertial Navigation

Given that the aircraft is a large-size target, it's inadequate to treat it as a mass point during positioning. To precisely align the aircraft position data acquired via DGPS with that from high-speed photography, it's imperative to convert the position of the target corner point to the DGPS antenna position. This conversion process can be rectified by leveraging aircraft attitude data gathered by inertial guidance systems.

The relationship between the coordinates of the DGPS antenna position and those of the target corner point is outlined as follows:

$$\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{bmatrix} \begin{bmatrix} \Delta x_a \\ \Delta y_a \\ \Delta z_a \end{bmatrix} \quad (13)$$

Where α, β, θ are the three-axis attitude angles of the aircraft's pitch, roll and yaw angles, $[\Delta x, \Delta y, \Delta z]^T$ is the coordinate difference between the DGPS antenna position and the target

point in the camera coordinate system, and $[\Delta x_a, \Delta y_a, \Delta z_a]^T$ is the coordinate difference between the DGPS antenna position and the target point in the airframe coordinate system.

Based on the coordinates of the fuselage target within the camera coordinate system, the coordinates of the DGPS antenna within the same coordinate system can be determined. Subsequently, these coordinates are converted into the geocentric coordinate system using the transformation relationship between the local coordinate system and the geocentric coordinate system. Finally, the results obtained from the high-speed camera method and the DGPS measurement method are compared.

5 Results

To validate the accuracy and robustness of the coordinate conversion, measurement methods, and image processing algorithms discussed in this paper, as well as to assess the measurement errors of the two methods—DGPS and monocular high-speed photography—three test flights were conducted at an airport. The test scenarios included flyovers, cross-sides, approaches, and level flights, conducted at altitudes of 1400ft, 1200ft, 1000ft, and 400ft. A total of 10 effective test points were executed. For the altitude conditions of 1200ft and 400ft, each involving a flight, the results are depicted in Figures 6 and 7. It can be observed that the

deviation remains within 3m for most points, although some exhibit significant deviations. The primary source of deviation stems from factors such as the accuracy of image recognition, lens distortion, and other related variables brought about by lighting conditions.

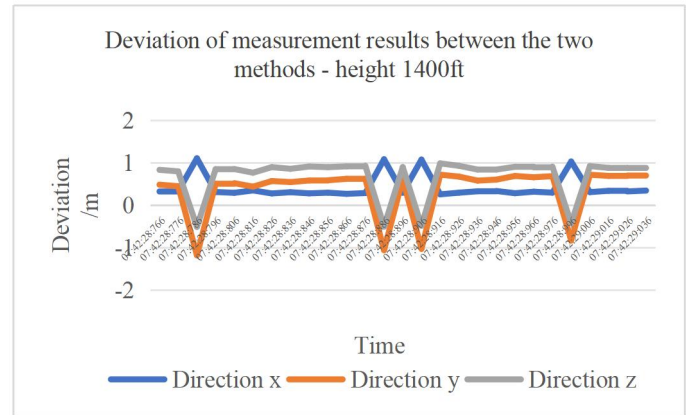


Fig. 8 Deviation of measurement results between the two methods for a flight at 1400ft altitude.

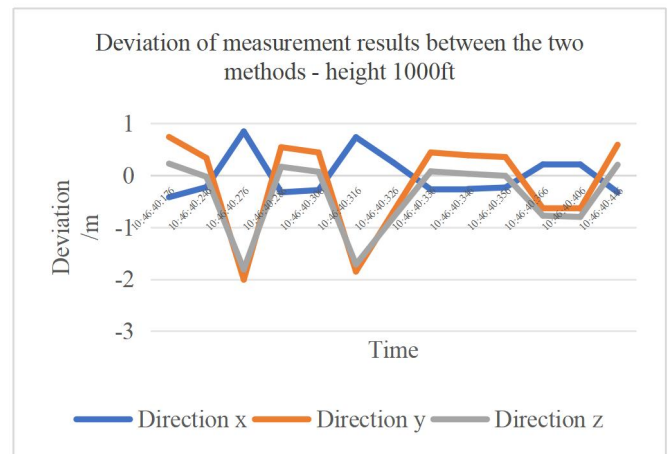


Fig. 9 Deviation of measurement results between the two methods for a flight at 1000ft altitude.

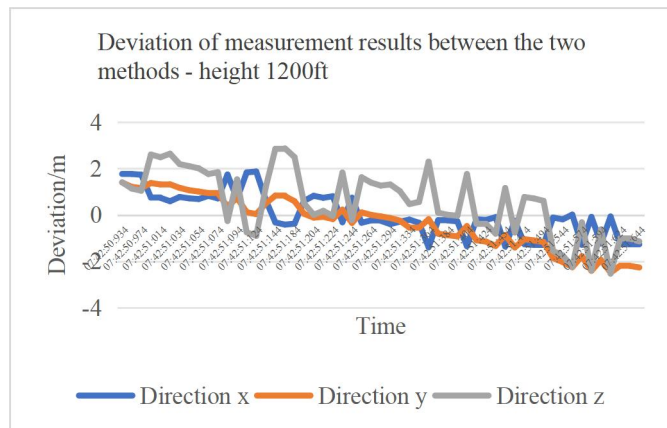


Fig. 6 Deviation of measurement results between the two methods for a flight at 1200ft altitude.

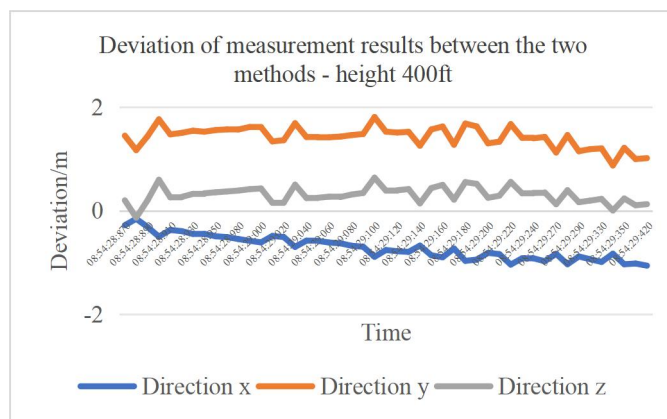


Fig. 7 Deviation of measurement results between the two methods for a flight at 400ft altitude.

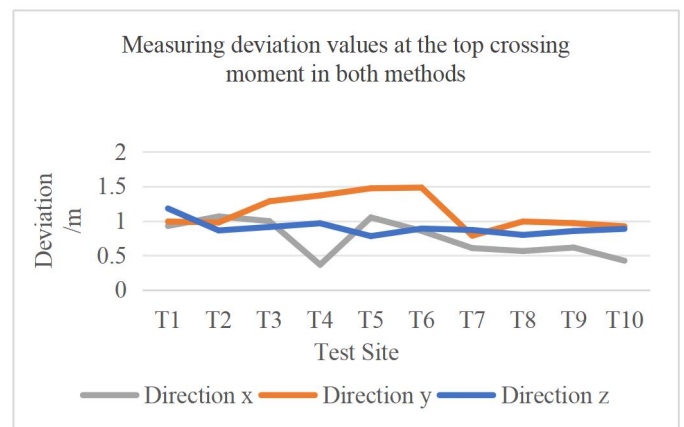


Fig. 10 Deviation values of DGPS and high-speed camera method results in three directions at the moment of camera head at different test points of overflight altitude.

Analyse the data of 10 valid test points, select the moment

when the aircraft is over the top, and compare the deviation of the measurement results of DGPS and high-speed camera method, as shown in Fig. 10, the deviation of the results of this moment are within 1.5 m.

The test data collected at various flight altitudes were analysed to compare the minimum root mean square (RMSE) of the deviation of measurements obtained from both the DGPS and high-speed camera methods. The statistical results are presented in Table 1, indicating that the RMSE in all directions remains within a deviation of 1.5m.

Table 1 Data acquisition of errors

test site	altitude	Δx	Δy	Δz
T1	1200ft	0.926m	0.991m	1.180m
T2	1200ft	1.064m	0.977m	0.861m
T3	1200ft	0.996m	1.284m	0.911m
T4	1400ft	0.363m	1.369m	0.965m
T5	400ft	1.050m	1.473m	0.779m
T6	1000ft	0.854m	1.483m	0.888m
T7	1400ft	0.605m	0.785m	0.868m
T8	1000ft	0.562m	0.991m	0.796m
T9	400ft	0.613m	0.967m	0.853m
T10	1000ft	0.422m	0.921m	0.884m

6 Conclusions

This paper highlights the feasibility and robustness of both the photogrammetric method utilizing a monocular photographic model and the correction method employing inertial guidance data. Through a comparison with DGPS results, it is demonstrated that the trajectory measurements

from both methods align closely, with deviations within 1.5 meters. This validation confirms the reliability of DGPS trajectory measurements, meeting the stipulations outlined in section 3.2.1.3.4 of ICAO DOC 9501 Volume I. This validation serves as a basis for utilizing monocular photography in trajectory measurement, providing valuable insights for its future application.

7 References

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